

Name: K RAJ BHARATH K RAJ BHARATH | **Reg. No:** RA2511003011548

Unit – 4 : Session – 6 : SLO – 2 | Hardwired Control — Fill in the Blanks

■ I. Fill in the Blanks

1. combinational logic (digital logic) — control signals are generated by gate-level circuits.
2. decoder — the instruction decoder decodes the opcode to initiate the appropriate control signals.
3. control memory — hardwired control eliminates control-memory access time, making it faster than microprogrammed control.
4. timing (sequence/order of steps) — the sequence counter determines which step of the instruction cycle is active.
5. combinational — signals are produced by combinational logic design (with sequential elements for state).
6. control logic (instruction set) — a major disadvantage is that the wired logic is difficult to modify.
7. sequence counter — keeps track of the current step in instruction execution.
8. RISC (simple, fixed) — hardwired control suits simple, fixed instruction set architectures.
9. clock (timing signal generator) — timing signals are generated by the clock and step logic.
10. logic gates — the entire control unit is built from logic gates and flip-flops.